

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error
from sklearn.metrics import mean_squared_error
from sklearn.metrics import r2_score

```

```
df = pd.read_csv("body_weight_dataset.csv")
```

```
df
```

	Weight	Height	Age	Gender
0	67.5	1.50	41	Male
1	61.9	1.58	31	Female
2	99.8	1.95	58	Male
3	73.0	1.72	46	Female
4	92.9	1.89	41	Male
..	...	...	...	...
195	81.1	1.77	52	Male
196	69.1	1.67	26	Male
197	70.4	1.69	53	Female
198	76.6	1.79	24	Female
199	80.5	1.75	40	Male

```
[200 rows x 4 columns]
```

```
df.head()
```

	Weight	Height	Age	Gender
0	67.5	1.50	41	Male
1	61.9	1.58	31	Female
2	99.8	1.95	58	Male
3	73.0	1.72	46	Female
4	92.9	1.89	41	Male

```
df.tail()
```

	Weight	Height	Age	Gender
195	81.1	1.77	52	Male
196	69.1	1.67	26	Male
197	70.4	1.69	53	Female
198	76.6	1.79	24	Female
199	80.5	1.75	40	Male

```
df.shape
```

```
(200, 4)
```

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 4 columns):
#   Column   Non-Null Count  Dtype
---  -
0   Weight   200 non-null    float64
1   Height   200 non-null    float64
2   Age       200 non-null    int64
3   Gender    200 non-null    str
dtypes: float64(2), int64(1), str(1)
memory usage: 6.4 KB
```

```
df.isnull().sum()
```

```
Weight    0
Height    0
Age        0
Gender     0
dtype: int64
```

```
df.describe()
```

	Weight	Height	Age
count	200.00000	200.000000	200.0000
mean	74.59600	1.697600	42.2350
std	11.09083	0.141294	13.7135
min	50.50000	1.450000	18.0000
25%	66.47500	1.580000	31.0000
50%	74.95000	1.690000	42.5000
75%	83.50000	1.810000	55.0000
max	99.80000	1.950000	65.0000

```
df.duplicated().sum()
```

```
np.int64(1)
```

```
df["Gender"] = df["Gender"].map({
    "Male":1,
    "Female":0
})
```

```
df.head()
```

	Weight	Height	Age	Gender
0	67.5	1.50	41	1
1	61.9	1.58	31	0
2	99.8	1.95	58	1
3	73.0	1.72	46	0
4	92.9	1.89	41	1

```
df.corr()
```

```

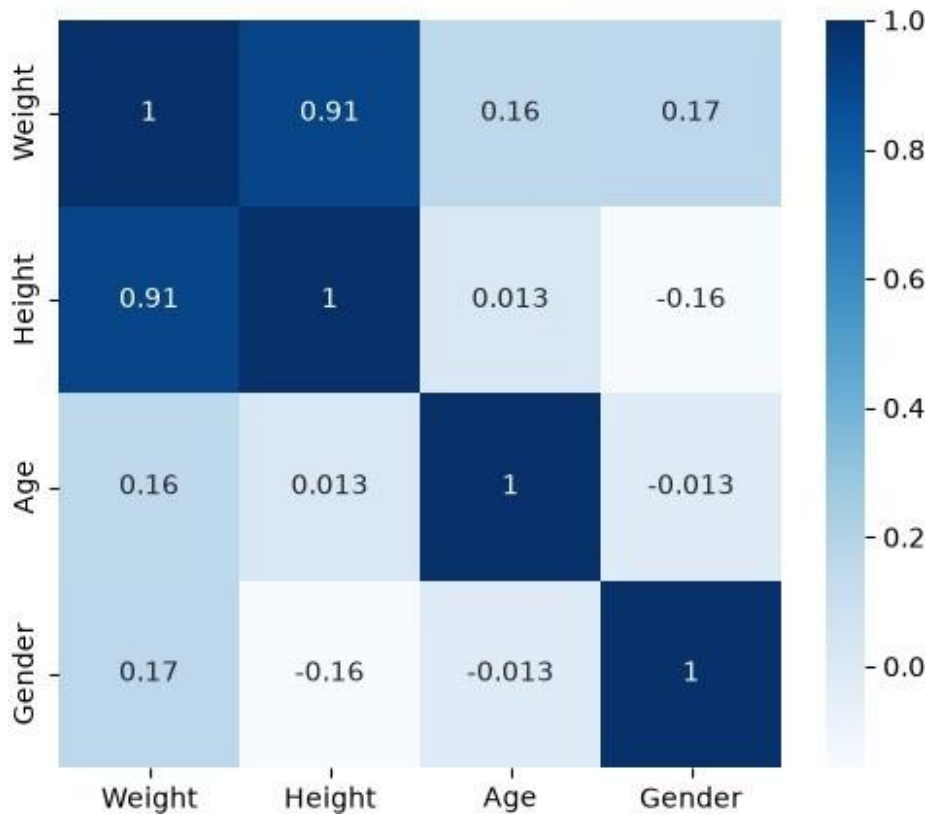
      Weight      Height      Age      Gender
Weight  1.000000  0.911290  0.162013  0.166781
Height  0.911290  1.000000  0.012974 -0.156432
Age      0.162013  0.012974  1.000000 -0.013413
Gender   0.166781 -0.156432 -0.013413  1.000000

```

```

plt.figure(figsize=(6,5))
sns.heatmap(df.corr(),annot=True,cmap="Blues")
plt.show()

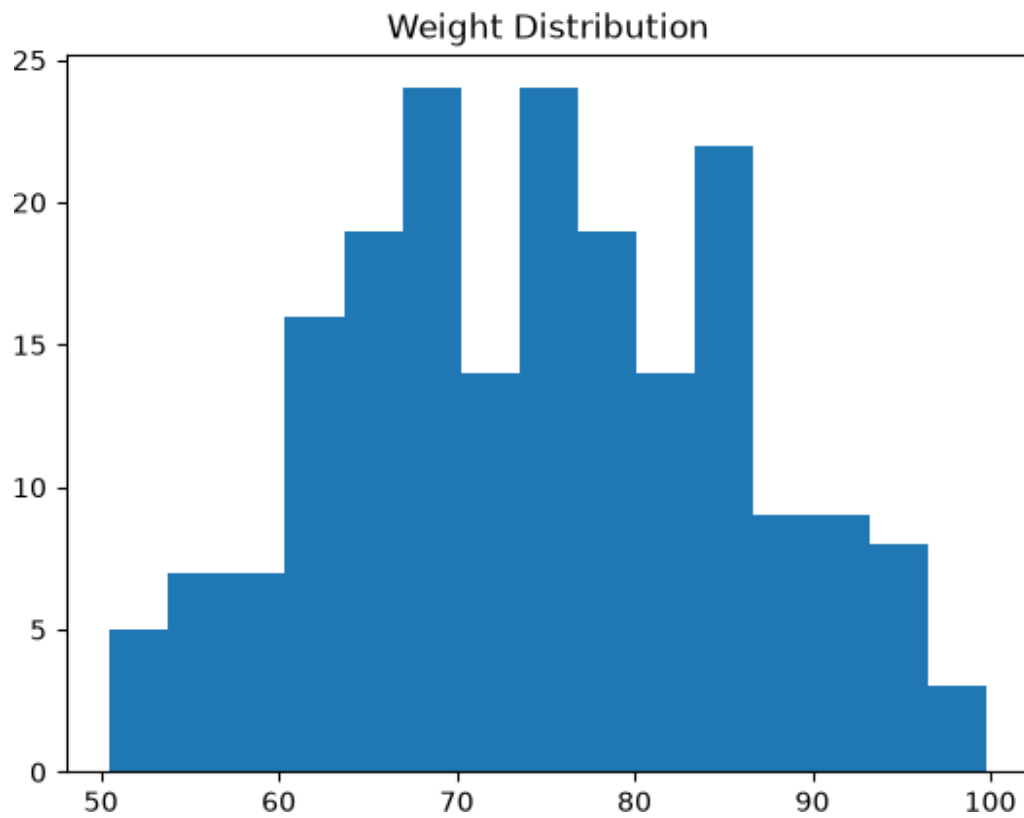
```



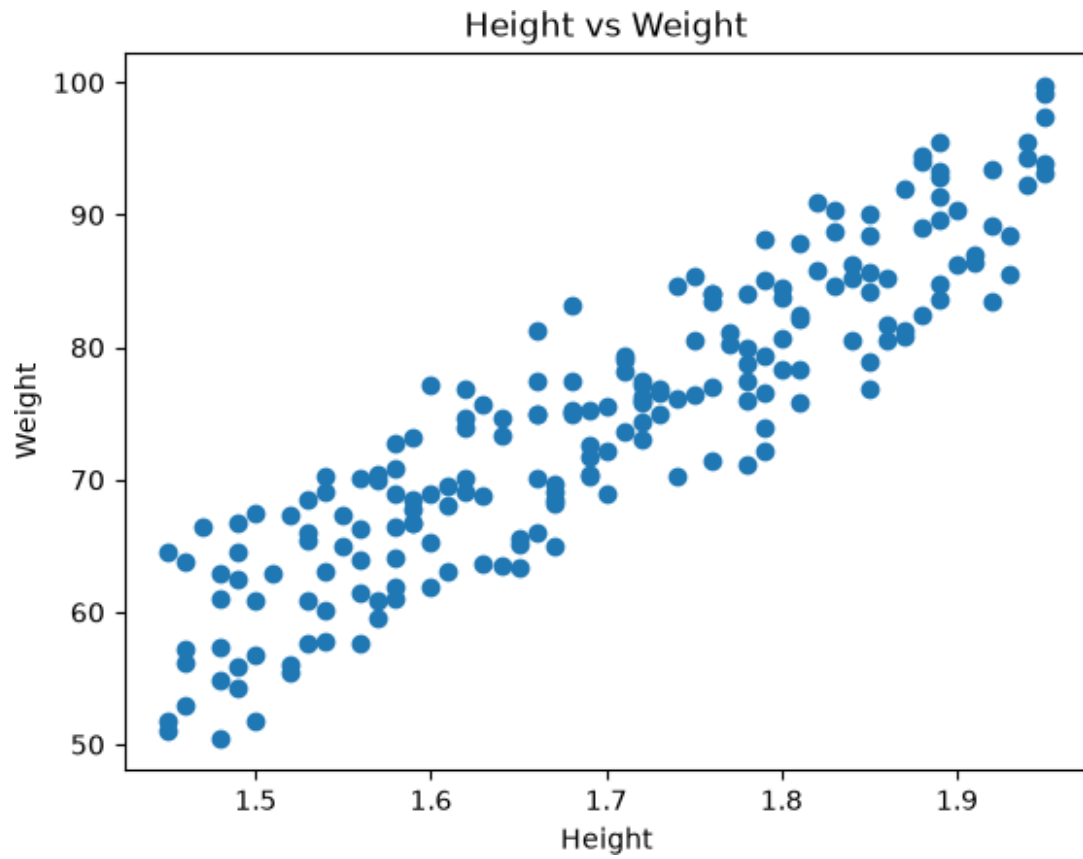
```

plt.hist(df["Weight"],bins=15)
plt.title("Weight Distribution")
plt.show()

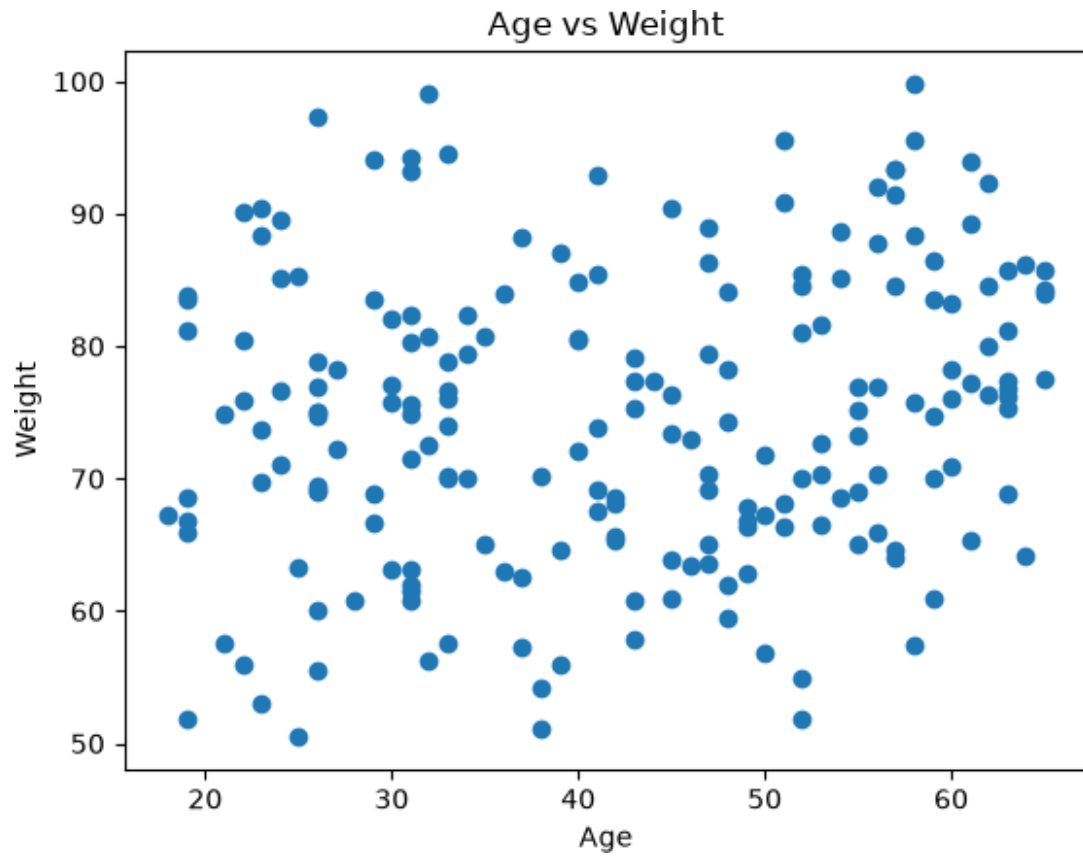
```



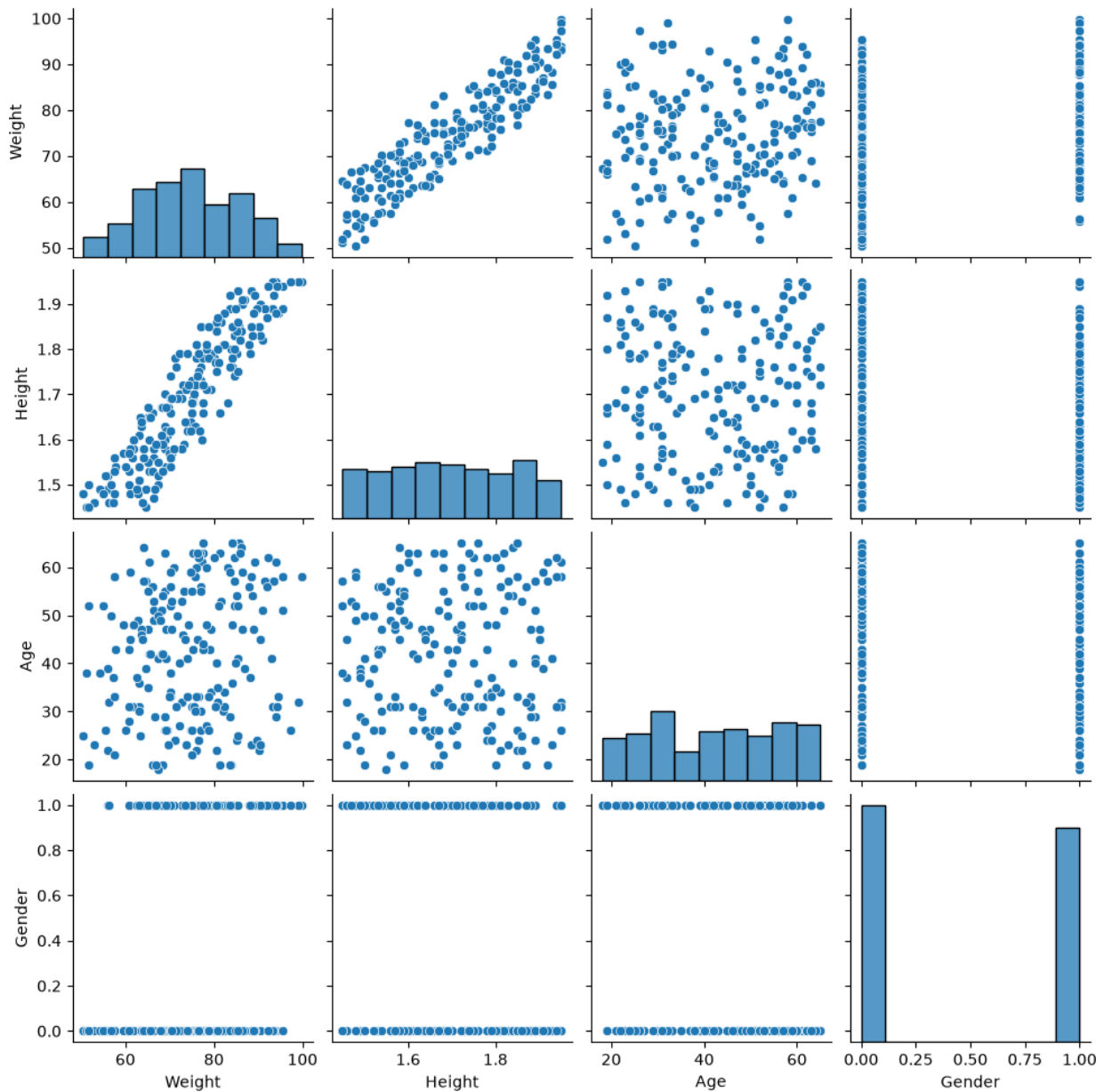
```
plt.scatter(df["Height"],df["Weight"])
plt.xlabel("Height")
plt.ylabel("Weight")
plt.title("Height vs Weight")
plt.show()
```



```
plt.scatter(df["Age"],df["Weight"])
plt.xlabel("Age")
plt.ylabel("Weight")
plt.title("Age vs Weight")
plt.show()
```



```
sns.pairplot(df)
plt.show()
```



```
X = df[["Height", "Age", "Gender"]]
```

```
y = df["Weight"]
```

```
X.head()
```

	Height	Age	Gender
0	1.50	41	1
1	1.58	31	0
2	1.95	58	1
3	1.72	46	0
4	1.89	41	1

```

y.head()
0    67.5
1    61.9
2    99.8
3    73.0
4    92.9
Name: Weight, dtype: float64

X_train,X_test,y_train,y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)

print(X_train.shape)
print(X_test.shape)

(160, 3)
(40, 3)

model = LinearRegression()
model.fit(X_train,y_train)
LinearRegression()

y_pred = model.predict(X_test)
print("MAE =",mean_absolute_error(y_test,y_pred))

MAE = 2.0998709048547246

mse = mean_squared_error(y_test,y_pred)
print("MSE =",mse)

MSE = 6.329184781005528

print("RMSE =",np.sqrt(mse))

RMSE = 2.51578710963498

print("R2 Score =",r2_score(y_test,y_pred))

R2 Score = 0.9552151618608655

print(model.coef_)

[76.17577728  0.11976253  7.28333164]

print(model.intercept_)

-63.216848278290186

```

```
person = pd.DataFrame({
    "Height": [1.75],
    "Age": [25],
    "Gender": [1]          # Male=1, Female=0
})

prediction = model.predict(person)

print("Predicted Weight =", prediction[0], "kg")

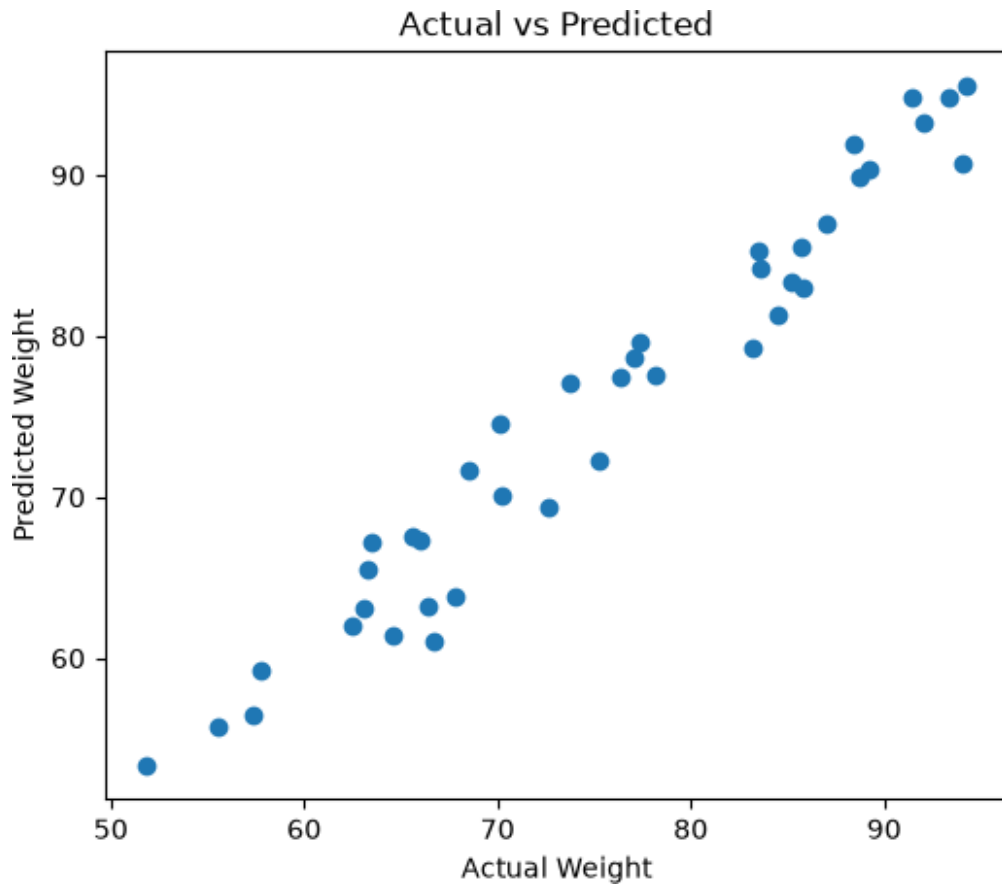
Predicted Weight = 80.3681568945941 kg

results = pd.DataFrame({
    "Actual": y_test,
    "Predicted": y_pred
})

results.head()

   Actual  Predicted
95    57.8    59.243638
15    72.6    69.352616
30    64.6    61.347825
158   92.0    93.221889
128   94.1    90.750058

plt.figure(figsize=(6,5))
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Weight")
plt.ylabel("Predicted Weight")
plt.title("Actual vs Predicted")
plt.show()
```



```
results.to_csv("BodyWeightPrediction.csv",index=False)
```

```
print("Prediction file saved successfully.")
```

```
Prediction file saved successfully.
```